Transport Obstruction G

1 Definition

The transport obstruction G is a mathematical quantity that measures the failure of a variable elliptic structure to be transported coherently with respect to its own multiplication law. Formally, it is defined as:

$$G := i_x + i_y, \tag{1}$$

where i_x and i_y are the derivatives of the generator $i(x, y)$ associated with the elliptic structure. The obstruction G vanishes identically when the structure is termed *rigid*, which is expressed as:

$$i_x + i_y = 0 \quad \text{on } \Omega. \tag{2}$$

This condition is intrinsic and purely geometric, indicating that the generator $i(x, y)$ behaves holomorphically with respect to its own variable elliptic structure.

2 Examples

Consider a variable elliptic structure defined by the polynomial:

$$X^2 + \beta(x, y)X + \alpha(x, y) = 0,$$

where α and β are functions of x and y. The generator $i(x, y)$ satisfies the structure relation:

$$i^2 + \beta i + \alpha = 0.$$

To compute the obstruction G, we differentiate i with respect to x and y:

1. Compute i_x and i_y using implicit differentiation of the structure relation. 2. Substitute these derivatives into the expression for G.

If, for example, we find that $G \equiv 0$ in a certain region Ω, this indicates that the structure is rigid in that region, allowing for a coherent analytic function theory to emerge.

3 Key Results

The concept of transport obstruction G is pivotal in understanding the dynamics of variable elliptic structures. Notable findings include:

- **Rigidity Condition**: A variable elliptic structure is rigid if $G \equiv 0$, which corresponds to the conservative case of the Burgers transport law $\lambda_x + \lambda\lambda_y = 0$ (Alayón-Solarz, 2026) [**?**]. - **Beltrami Coefficient Relation**: The rigidity condition can be expressed in terms of the Beltrami coefficient μ as $\overline{\mu}_z = \mu\mu_z$ (Theorem 12.1, Alayón-Solarz, 2026) [**?**]. - **Normalized Obstruction**: The normalized obstruction ρ_T that controls rigidization convergence is given by:

$$\rho_T \leq 4\|R(\mu)\|_\infty \cdot \|1 - \mu\|_\infty \cdot (1 - \|\mu\|_\infty^2)^{-2},$$

where $R(\mu)$ is the Poincaré residual (Proposition 12.8, Alayón-Solarz, 2026) [**?**].

4 Diagrams / Visual Description

To visualize the concept of transport obstruction G, consider a diagram depicting the variable elliptic structure on the complex plane. The structure can be represented as a surface where the values of α and β vary. The obstruction G can be illustrated as a vector field on this surface, indicating regions of rigidity (where $G = 0$) and non-rigidity (where $G \neq 0$).

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5 Connections

The transport obstruction G is closely related to several concepts in the field of complex analysis and differential geometry:

- **Elliptic Systems**: G is directly tied to the behavior of elliptic systems, where it measures the compatibility of the system's structure with its differential equations. - **Beltrami Coefficients**: The relationship between G and the Beltrami coefficient μ provides a bridge between the theory of variable elliptic structures and classical hyperbolic geometry. - **Cauchy-Riemann Systems**: The presence of G influences the inhomogeneity of generalized Cauchy-Riemann systems, affecting the existence and uniqueness of solutions to these systems.

6 References

- Alayón-Solarz, D. (2026). Variable Elliptic Structures on the Plane: Transport Dynamics, Rigidity, and Function Theory. arXiv:2601.19274v4.